

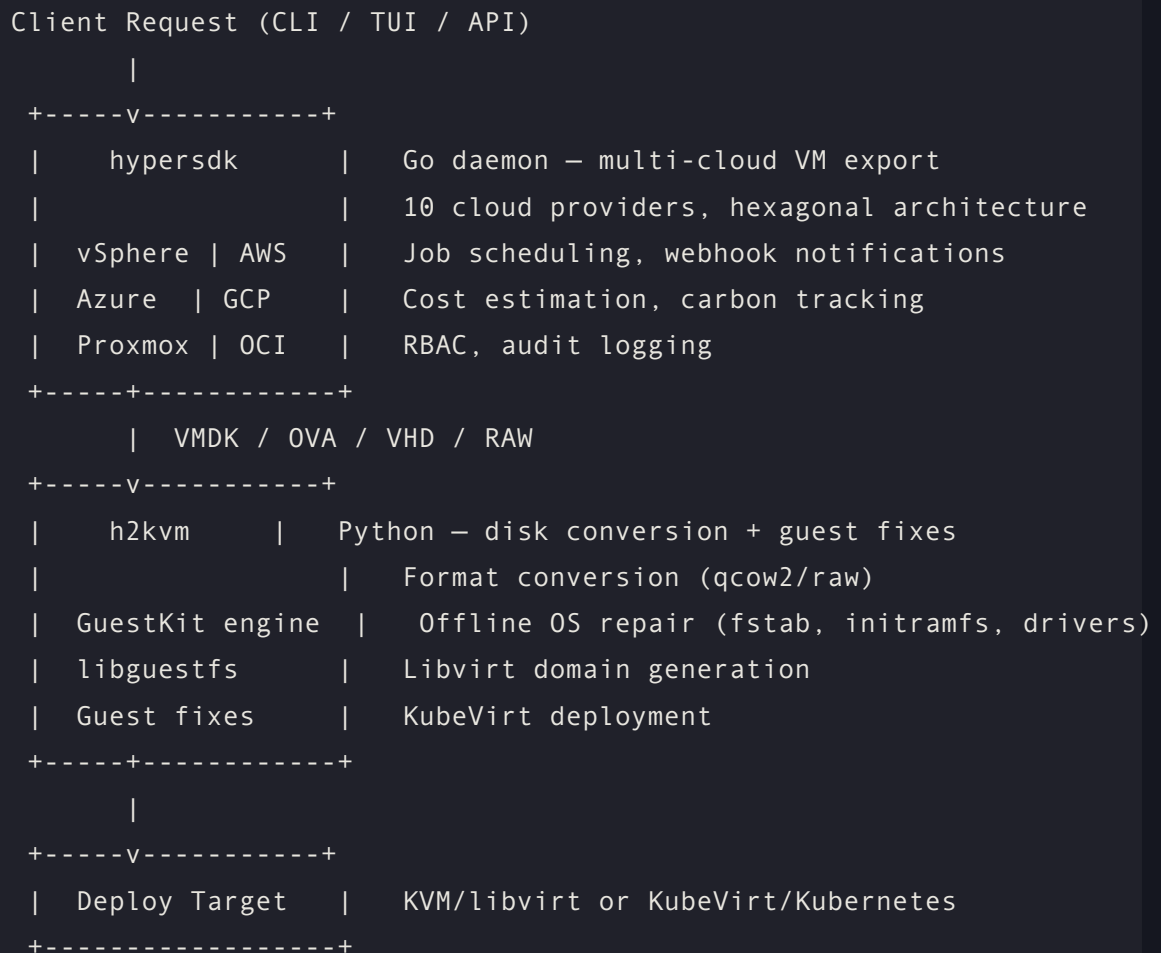
hyper2kvm + hypersdk

Technical Architecture Deep Dive

How the two-layer platform handles multi-cloud VM export, conversion, and deployment at enterprise scale.

Two-Layer Architecture

hypersdk handles cloud APIs. h2kvm handles disk conversion and guest fixes.



hypersdk (Go)

Cloud-native daemon with REST API. Handles provider authentication, VM discovery, disk export, job queue, scheduling. Hexagonal architecture with 13+ port interfaces.



h2kvm (Python)

Disk processing engine. Converts formats, applies offline guest fixes, generates libvirt XML, deploys to KubeVirt. 1,273 tests. Interactive TUI (zkvm).

hypersdk: 10 Cloud Providers

One API, every hypervisor and cloud platform.

Provider	Export Method	Auth	Features
VMware vSphere	OVF/OVA, HTTPS	SSO, username/password	CBT delta sync, snapshot export
AWS EC2	EBS snapshot, AMI export	IAM, access keys	Cross-region, S3 staging
Microsoft Azure	Managed disk export	Service principal, MSI	Snapshot-based, parallel download
Google Cloud	Persistent disk export	Service account	GCS staging, image export
Microsoft Hyper-V	VHD/VHDX direct	WinRM, SSH	Checkpoint export
Oracle Cloud (OCI)	Boot volume export	API key	Object storage staging
OpenStack	Glance image, Cinder	Keystone v3	Swift staging
Alibaba Cloud	ECS disk export	AccessKey	OSS staging
Proxmox VE	API disk export	Token, PAM	qcow2/raw direct
KubeVirt	PVC/DataVolume	kubeconfig	Live migration, CDI

Provider Interface (Go)

```
type Provider interface {
    ListVMs(ctx context.Context, filter VMFilter) ([]VMInfo, error)
    ExportVM(ctx context.Context, vmID string, opts ExportOptions) (*ExportResult, error)
    GetCapabilities() []ExportCapability
    Validate(ctx context.Context) error
}
```

hypersdk: Hexagonal Architecture

Clean separation of domain, ports, and adapters.



Domain Layer

70+ canonical Go types. Job, Schedule, VM, Cost, Carbon, Webhook, Audit, RBAC. Zero dependencies on external packages.



Ports Layer

13+ interfaces defining contracts. JobManager, SchedulerManager, CapabilityDetector, CostEstimator, CarbonTracker, AuditLogger.



Adapters Layer

Concrete implementations. REST API, provider clients, SQLite store, Prometheus metrics, webhook dispatcher, secrets vault.

Key Daemon Features

Job Scheduling

- Cron-based scheduling with time windows
- Retry policies with exponential backoff
- Priority queues, concurrency limits
- Webhook notifications on completion

Observability

- Prometheus metrics endpoint
- Structured JSON logging with trace IDs
- Audit trail for all operations
- Cost estimation per migration

Security

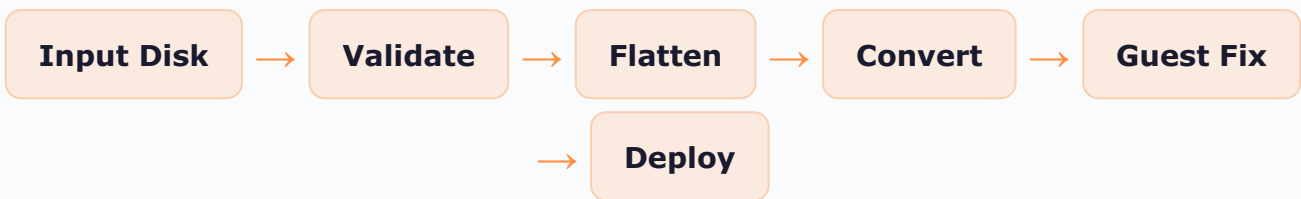
- RBAC with roles and permissions
- API key + JWT authentication
- Secrets management (vault integration)
- TLS everywhere, cert rotation

Enterprise Features

- Carbon-aware scheduling (green migration)
- Multi-tenant with namespace isolation
- Backup and disaster recovery
- HA with leader election

h2kvm: Conversion Pipeline

What happens to a VM disk from export to first boot.



Stage	What It Does	Why It Matters
Validate	VMDK descriptor parsing, extent verification, boot mode detection (BIOS/UEFI), risk analysis	Catches corrupt/truncated disks before wasting hours on conversion
Flatten	Merge snapshot chain into single image via qemu-img convert	Eliminates snapshot dependencies, produces self-contained disk
Convert	VMDK/VHD/RAW to qcow2 with optional compression	qcow2 is the native KVM format — thin provisioning, snapshots, compression
Guest Fix	Mount disk offline, fix fstab (UUID stabilization), rebuild initramfs with virtio drivers, remove VMware tools, fix GRUB, relabel SELinux	Without this, VMs fail to boot on KVM — missing drivers, wrong device paths
Deploy	Generate libvirt XML, virsh define, or create KubeVirt VirtualMachine + PVC upload	VM is ready to start — no manual XML editing or kubectl gymnastics

Guest Fix Engine (GuestKit)

Linux Fixes

- fstab: convert device names to UUID/path
- initramfs: inject virtio_blk, virtio_net, virtio_scsi
- GRUB: update root= parameter, rebuild config

Windows Fixes

- VirtIO driver injection (disk, net, balloon)
- Registry: update boot device driver stack
- Remove VMware Tools/Hyper-V IC services
- Fix BCD boot configuration

- SELinux: trigger autorelabel on next boot
- Network: remove VMware/Hyper-V NIC configs
- VMware tools: remove open-vm-tools packages

- Enable serial console for headless access
- Multi-stage: bootstrap, drivers, cleanup

zkvm: Interactive TUI

Terminal UI built with Go + Bubble Tea for guided migration.



Guided Steps

Step 1: Source (auto-detect format) → Step 2: Output → Step 3: Fixes → Step 4: Deploy to Libvirt. No CLI memorization.



vSphere Integration

Auto-populate from GOVC_* env vars. Discover VMs with CPU/RAM/power state. Select, queue, migrate — all from the TUI.



File Browser

Built-in fzf-style fuzzy file picker. Filters by extension (.vmdk, .ova, etc). Navigate dirs, search instantly.



Live Progress

Real-time streaming of govc export progress, conversion status, and deployment steps. Progress bar with speed and ETA.

5 Tabs

Tab	Purpose
1. Migration	Config form (guided steps) + deploy targets + execution plan
2. vSphere	vCenter connection, VM discovery, batch selection, queue migration
3. Logs	Full scrollable log viewer with color-coded levels
4. Libvirt VMs	List/manage local KVM VMs, start/stop/delete, console access
5. Kubernetes	KubeVirt VMs, PVCs, namespaces, multi-context support

Deployment Options

Flexible deployment for every environment.



Standalone CLI

Single binary + Python package. Run on any Linux box. YAML-driven for automation. Perfect for scripts and CI/CD.



Interactive TUI

zkvm terminal UI for guided migration. File browser, vSphere discovery, live progress. Zero learning curve.



Daemon Mode

hypersdk runs as systemd service. REST API for programmatic access. Job scheduling, webhooks, RBAC.



Kubernetes Operator

Deploy as K8s operator. CRDs for migration jobs. GitOps-friendly. Helm chart included.



Container Image

OCI container for Podman/Docker. Mount disks as volumes, convert in isolation. CI/CD pipeline ready.



RPM / DEB Package

Native packages for RHEL, Fedora, Ubuntu, Debian. Standard install, systemd integration, man pages.